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Assessment-2

ISWE406P - Agile Development Process and Devops Lab

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COURSE CODE: ISWE406P

JENKINS

1. Learning Goal

Understand **what Jenkins is** and run a job successfully.

Stepwise Tasks

1. What is CI/CD? (Real-world example)

1.Continuous Integration (CI)

Developers frequently push their code to a shared repository, and **every change is automatically checked**.

What happens in CI?

- Code is pushed (GitHub / GitLab / Bitbucket)
- Automatic build starts
- Automatic tests run
- Errors are caught early

Why CI is important:

- Finds bugs quickly
- Avoids “it works on my machine” problems
- Keeps code stable

2.Continuous Delivery / Deployment (CD)

Continuous Delivery

- Code is **always ready to release**
- Deployment is **manual** (you click a button)

Continuous Deployment

- Code is **automatically released to users**
- No manual approval needed

Real-World Example

Online Shopping App (like Amazon-style app)

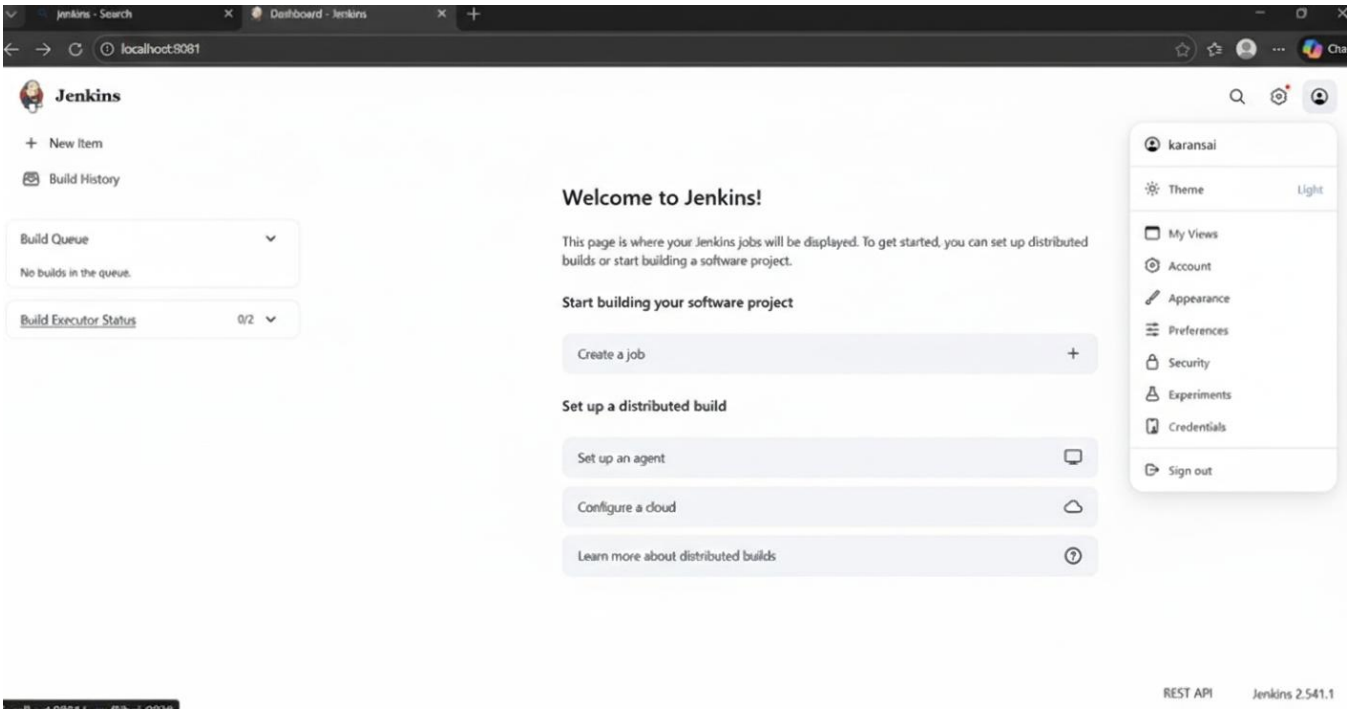
Without CI/CD (Old method)

- Developer changes payment code
- Tests manually on their PC
- Sends files to server
- App crashes after release
- Fix takes days

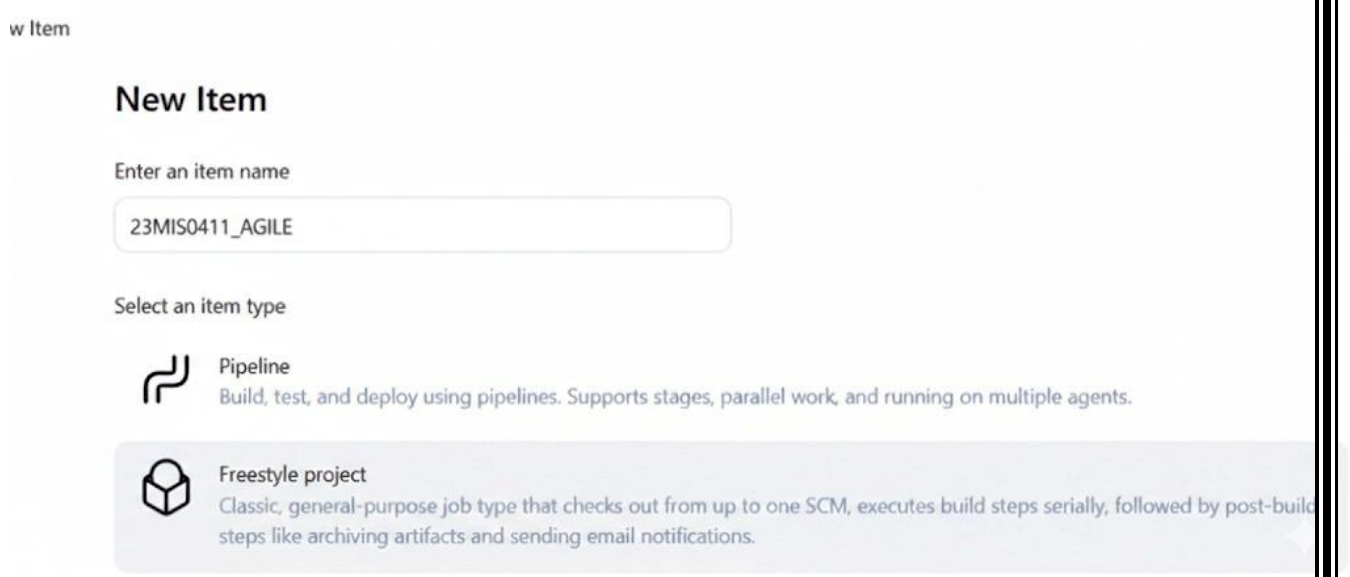
With CI/CD (Modern method)

1. **Developer updates code** (e.g., fix UPI payment bug)
2. **CI starts automatically**
    - App is built
    - Unit tests check payment logic
    - Security checks run
3. **If tests pass → CD starts**
    - App is deployed to staging
    - Then pushed to production

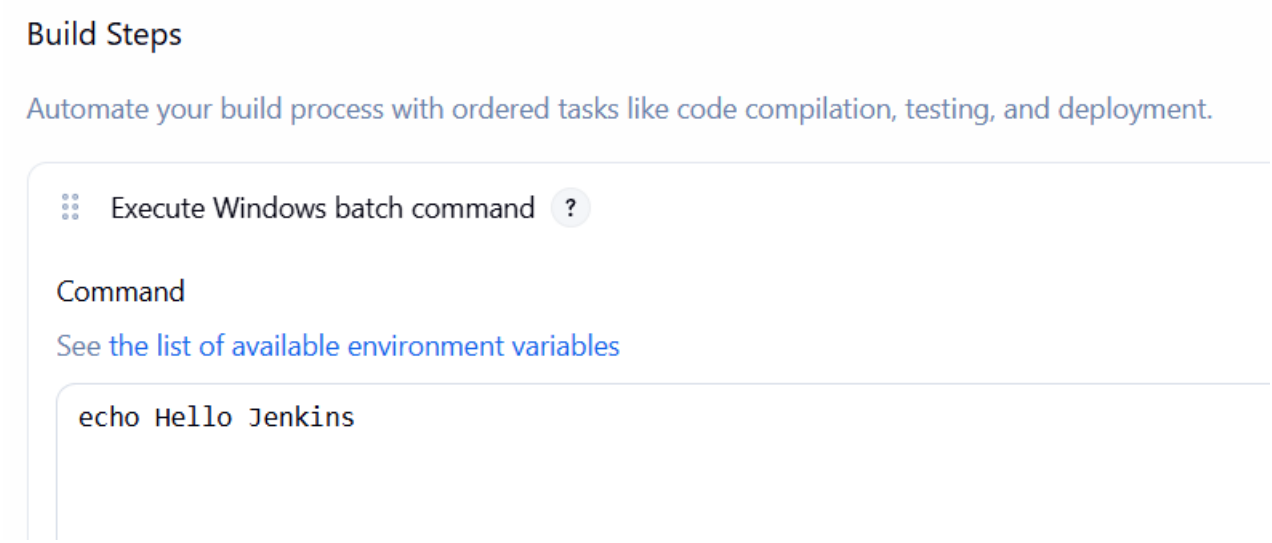
2. Install Jenkins (Windows / Linux)



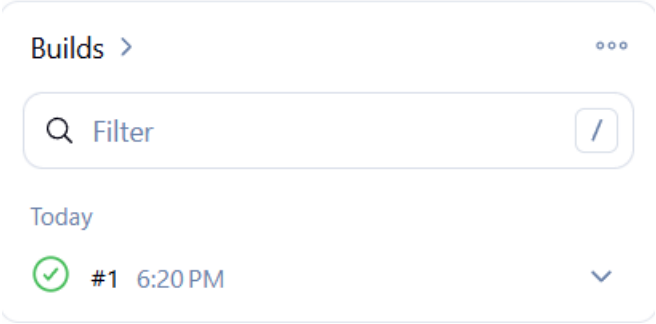
3. Jenkins UI overview
4. Create **Freestyle Job**



5. Add simple build step:
    - Windows: echo Hello Jenkins



6. Run job manually



7. View:

Console Output:

Console

```
Started by user karan326
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\23MIS0411_AGILE
[23MIS0411_AGILE] $ cmd /c call C:\WINDOWS\TEMP\jenkins3859394143266103641.bat

C:\ProgramData\Jenkins\.jenkins\workspace\23MIS0411_AGILE>echo Hello Jenkins
Hello Jenkins

C:\ProgramData\Jenkins\.jenkins\workspace\23MIS0411_AGILE>exit 0
Finished: SUCCESS
```

Build History:

Builds >

Filter /

Today

#2 6:26 PM

#1 6:20 PM

Build History of Jenkins

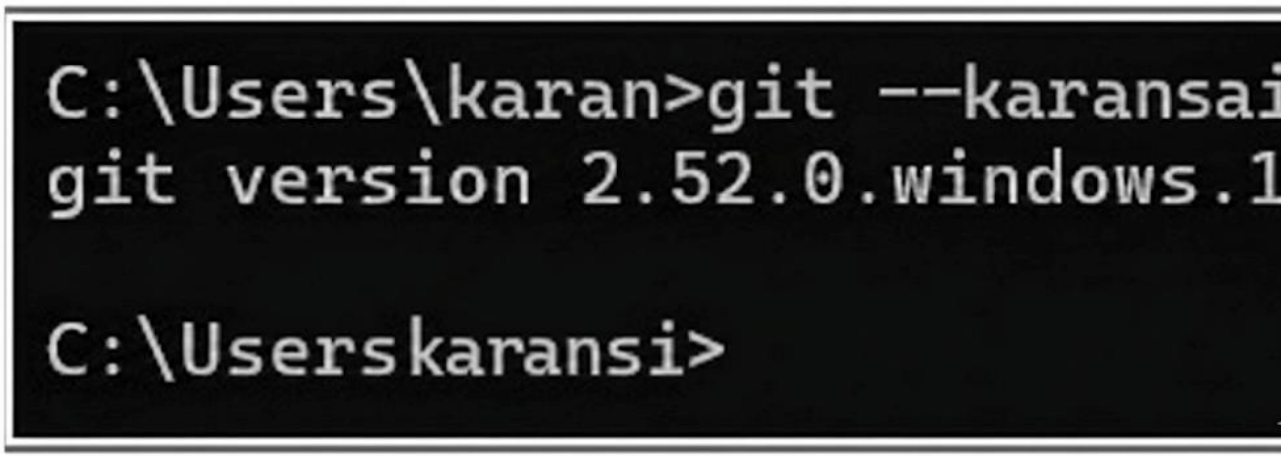
S	Build	Time Since ↑	Status
	3MIS0411_AGILE #2	40 min	stable
	23MIS0411_AGILE #1	46 min	stable

2. Learning Goal

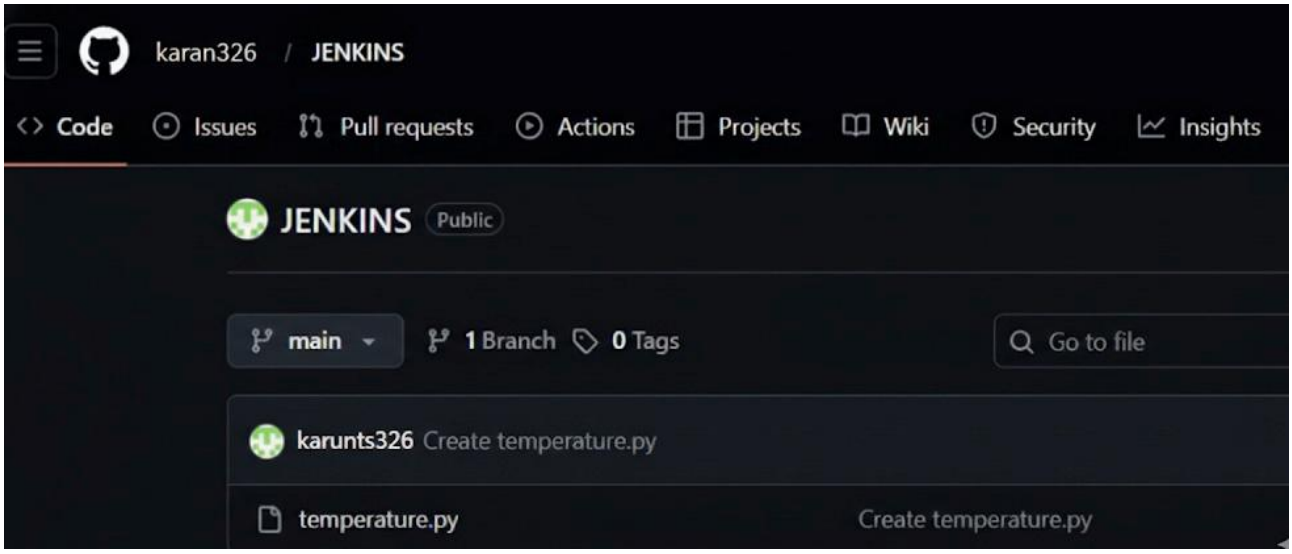
Work with **source code and build automation**

Stepwise Tasks

1. Install Git



2. Create GitHub repository



3. Configure **Git in Jenkins**

Git installations

Git

Name

AGILE

Path to Git executable ?

C:\Program Files\Git\bin\git.exe

4. Create Freestyle Job:

Source Code Management → Git:

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

None

Git ?

Repositories ?

Repository URL ?

https://github.com/karan326/JENKINS.git

5. Add build step:

Windows batch / Shell

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

```
python temperature.py
```

6. Poll SCM / Webhook trigger

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- [ ] Trigger builds remotely (e.g., from scripts) ?
- [ ] Build after other projects are built ?
- [ ] Build periodically ?
- [ ] GitHub hook trigger for GITScm polling ?
- [x] Poll SCM ?

Schedule ?

```
H/2 * * * *
```

7. Build automatically on code change

Builds >

Filter /

Today

#1 7:37 PM

AGILE_GITCLONE

Permalinks

- Last build (#1), 5 min 27 sec ago
- Last stable build (#1), 5 min 27 sec ago
- Last successful build (#1), 5 min 27 sec ago
- Last completed build (#1), 5 min 27 sec ago

Console

Download Copy View as plain tex

```
Started by user Karanasi
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\AGILE_GITCLONE
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/arrunts491/JENKINS.git
 > C:\Program Files\Git\bin\git.exe init C:\ProgramData\jenkins\.jenkins\workspace\AGILE_GITCLONE # timeout=10
Fetching upstream changes from https://github.com/arrunts491/JENKINS.git
 > C:\Program Files\Git\bin\git.exe --version # timeout=26
 > git --version # 'git version 2.22.0.windows.1'
 > C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/arrunts491/JENKINS.git +refs/heads/*:refs/remotes/origin/* # timeout=10
 > C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/arrunts491/JENKINS.git # timeout=10
 > C:\Program Files\Git\bin\git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/main*(commit)" # timeout=10
Checking out Revision de002f520283330d3e8214de45b4a269f2e7f4e7 (refs/remotes/origin/main)
 > C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
 > C:\Program Files\Git\bin\git.exe checkout -f de002F528283238d3e8214de45b4a269f2e7f4e7 # timeout=10
Commit message: "Create temperature.py"
First time build. Skipping changeing.
[AGILE_GITCLONE] $ end /c call C:\WINDOWS\TEMP\jenkins13399527871760200398.bat

C:\ProgramData\Jenkins\.jenkins\workspace\AGILE_GITCLONE>echo Git Build Success
Git Build Success

C:\ProgramData\Jenkins\.jenkins\workspace\AGILE_GITCLONE>exit 0
Finished: SUCCESS
```

3. Learning Goal

Understand **job flow visually**

Stepwise Tasks

1. Create multiple Freestyle jobs:
    - Job-A (Build)
    - Job-B (Test)
    - Job-C (Deploy)

		JOB-A-BUILD	N/A	N/A	N/A
		JOB-B-TEST	N/A	N/A	N/A
		JOB-C-DEPLOY	N/A	N/A	N/A

1. Use:
    - Post-build action → Trigger other projects

FROM JOB-A-BUILD:

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Build other projects ?

Projects to build

JOB-B-TEST

- (●) Trigger only if build is stable
- ( ) Trigger even if the build is unstable
- ( ) Trigger even if the build fails

+ Add post-build action

Save Apply

FROM JOB-B-TEST:

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Build other projects ?

Projects to build

JOB-C-DEPLOY

- (●) Trigger only if build is stable
- ( ) Trigger even if the build is unstable
- ( ) Trigger even if the build fails

3.View job chaining visually

Jenkins / JOB-A-BUILD / #1

Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings

Console

Download | Copy | View as plain text

```
Started by user Karansai
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\JOB-A-BUILD
[JOB-A-BUILD] $ cmd /c call C:\WINDOWS\TEMP\jenkins16466338558352965343.bat

C:\ProgramData\Jenkins\.jenkins\workspace\JOB-A-BUILD>echo BUILD STAGE
BUILD STAGE

C:\ProgramData\Jenkins\.jenkins\workspace\JOB-A-BUILD>exit 0
Triggering a new build of JOB-B-TEST
Finished: SUCCESS
```

Jenkins / JOB-C-DEPLOY / #1

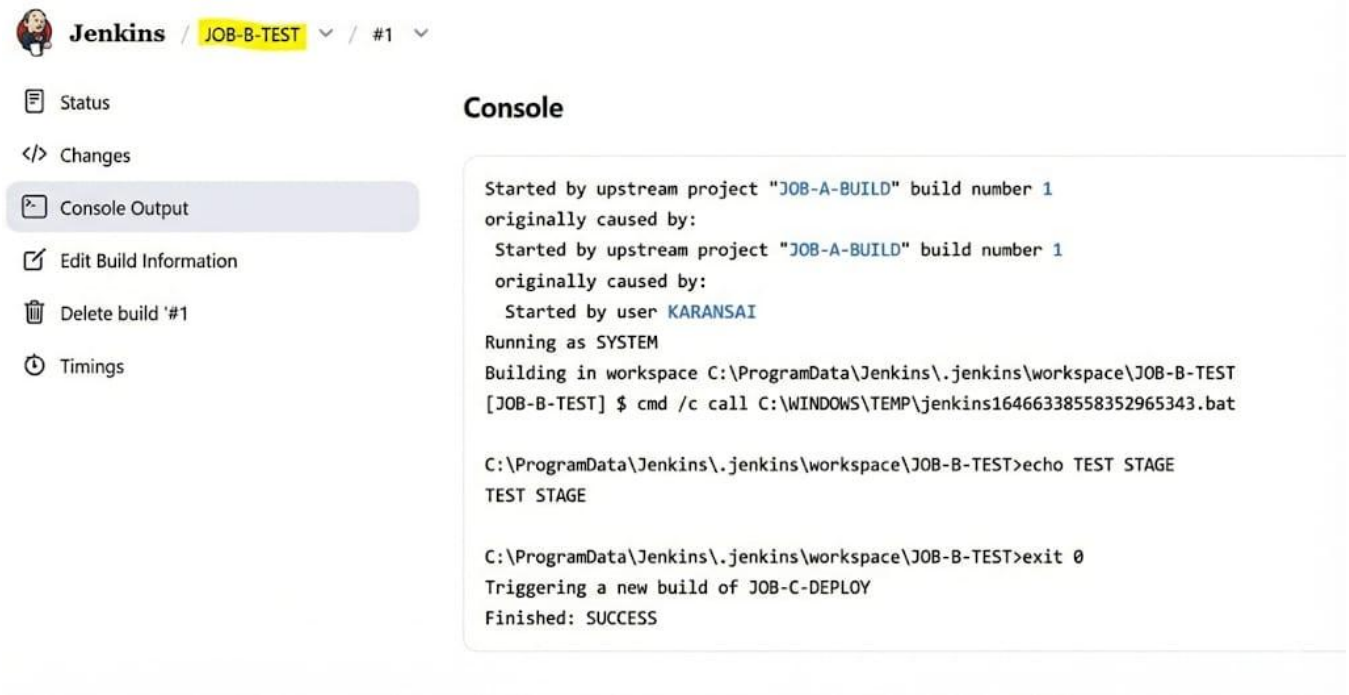
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings

Console

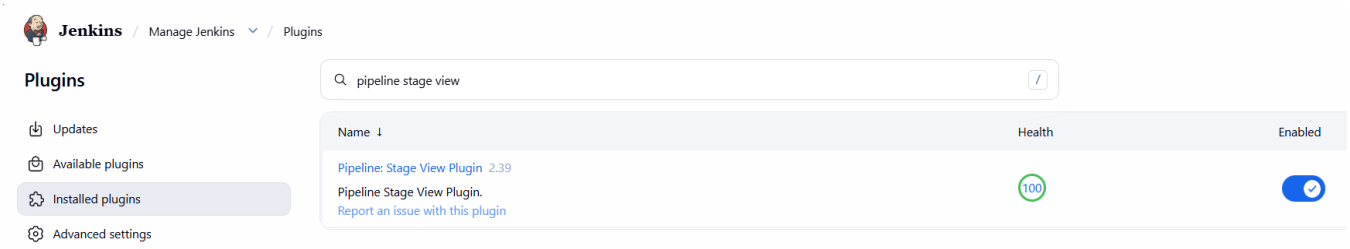
```
Started by upstream project "JOB-B-TEST" build number 1
originally caused by:
 Started by upstream project "JOB-A-BUILD" build number 1
 originally caused by:
  Started by user KARANSAI
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\JOB-C-DEPLOY
[JOB-C-DEPLOY] $ cmd /c call C:\WINDOWS\TEMP\jenkins16944590688656969345.bat

C:\ProgramData\Jenkins\.jenkins\workspace\JOB-C-DEPLOY>echo DEPLOY STAGE
DEPLOY STAGE

C:\ProgramData\Jenkins\.jenkins\workspace\JOB-C-DEPLOY>exit 0
Finished: SUCCESS
```



4.Install **Pipeline Stage View**



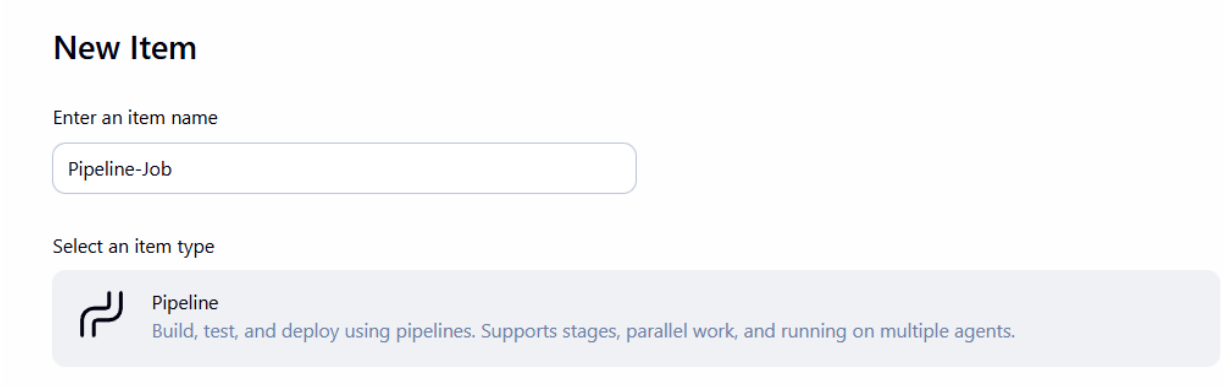
5.Observe stage-wise execution

4. Learning Goal

Use **Jenkins Pipeline & Jenkinsfile**

Stepwise Tasks

1. What is Jenkins Pipeline?
2. Scripted vs Declarative Pipeline
3. Create **Pipeline Job**



4. Write pipeline in Jenkins UI
5. Add stages:
    - Build
    - Test
    - Deploy
6. Introduce:
    - agent
    - stages
    - steps

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

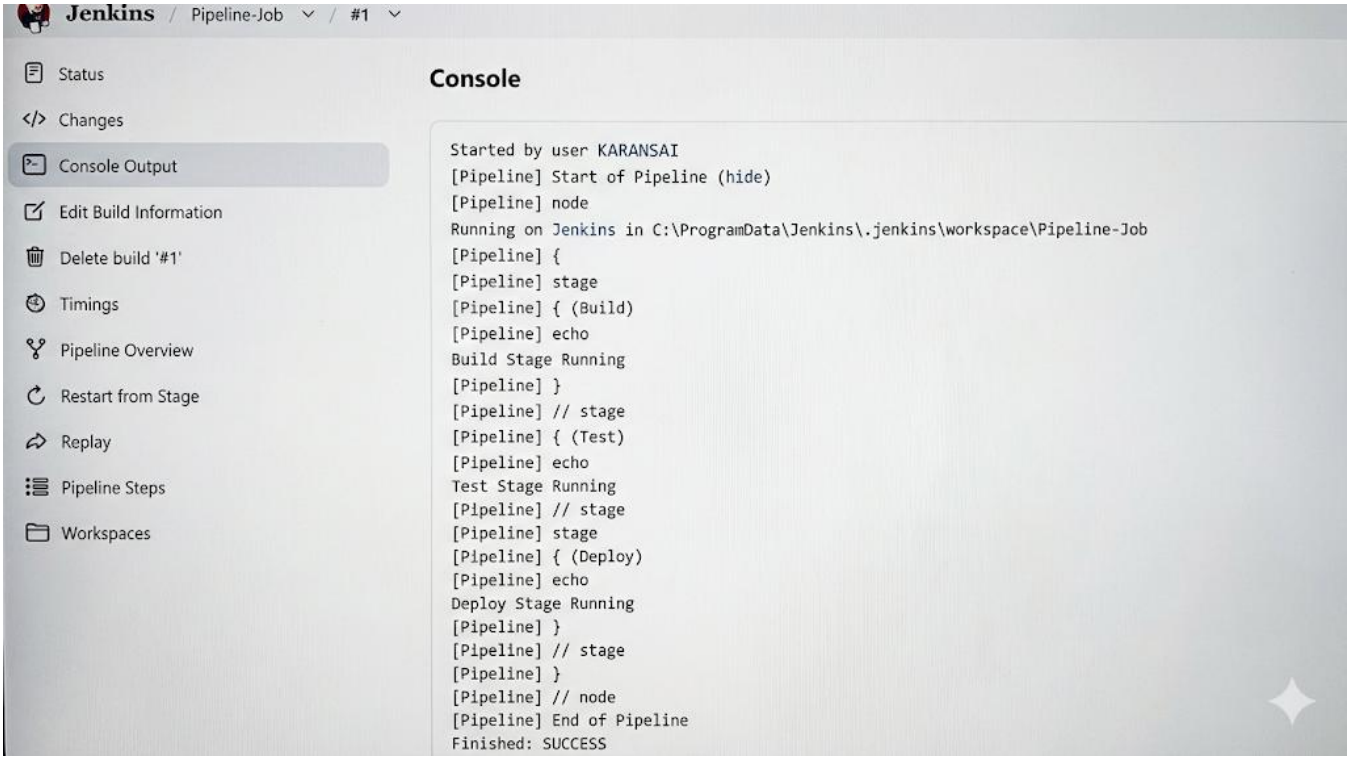
Pipeline script

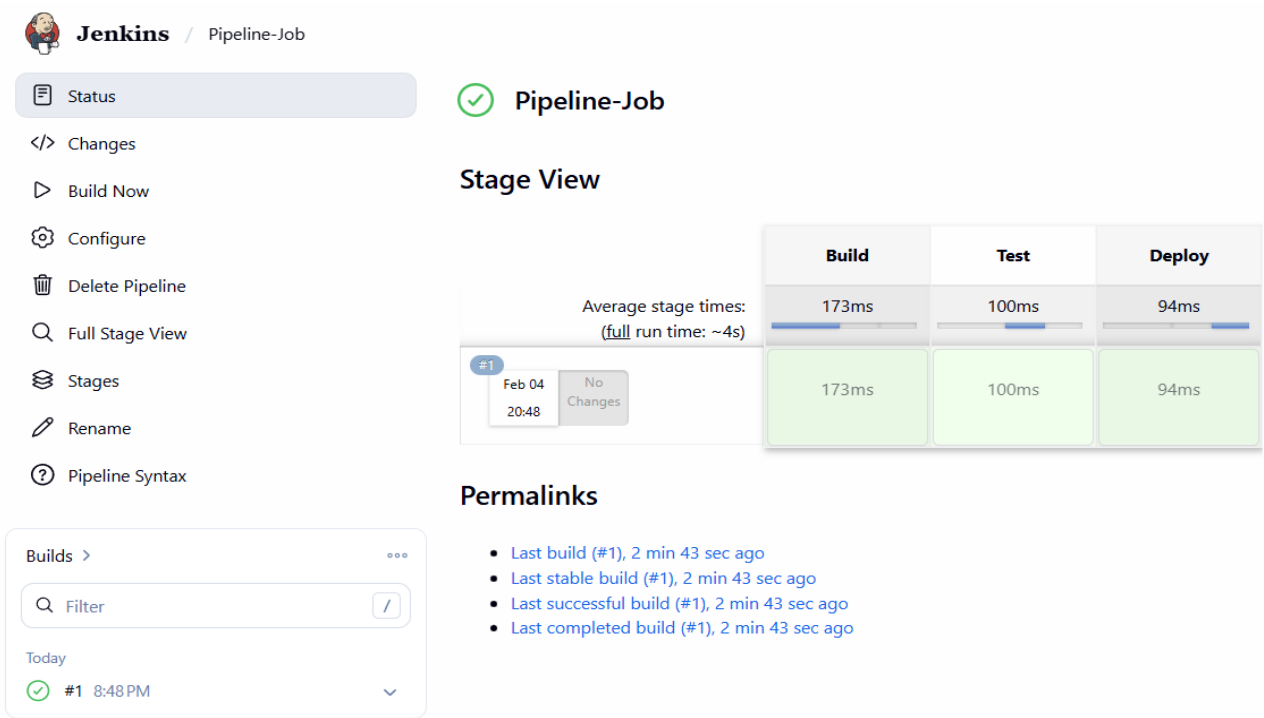
Script ?

```
pipeline {
  agent any
  stages {
    stage('Build') {
      steps {
        echo 'Build Stage Running'
      }
    }
    stage('Test') {
      steps {
        echo 'Test Stage Running'
      }
    }
    stage('Deploy') {
      steps {
```

```
    stage('Deploy') {
      steps {
        echo 'Deploy Stage Running'
      }
    }
  }
}

```



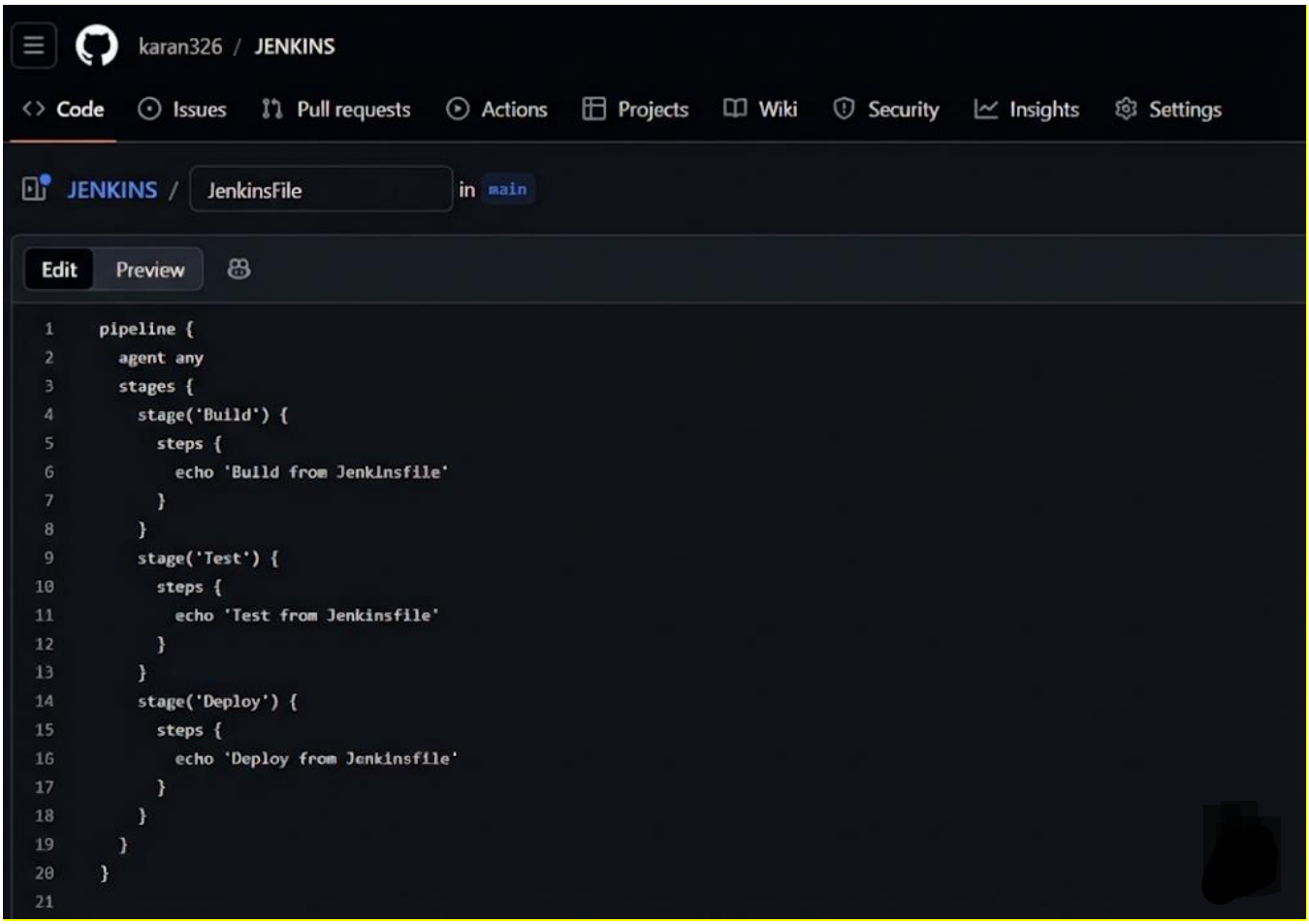


5. Learning Goal

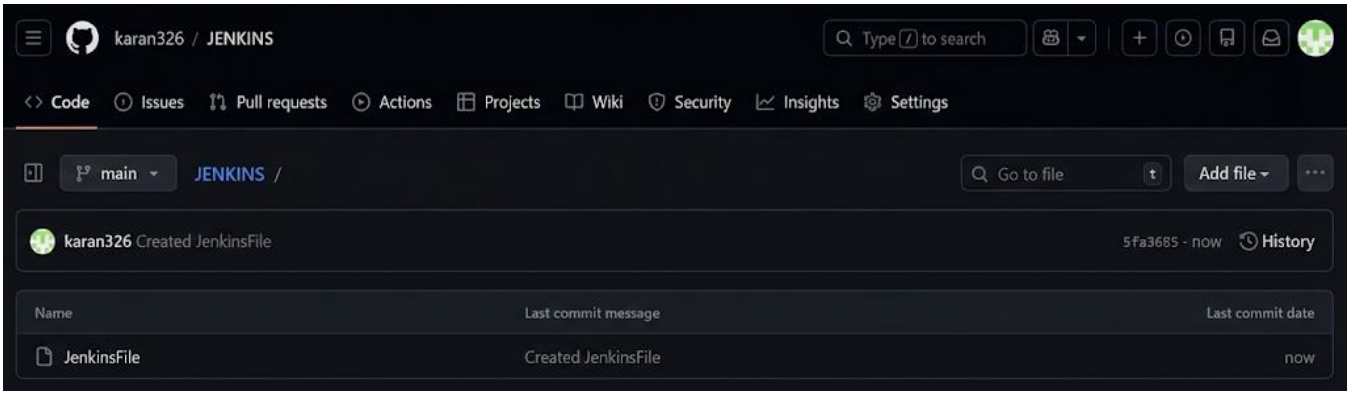
Pipeline as code from repository

Stepwise Tasks

1.Create Jenkinsfile

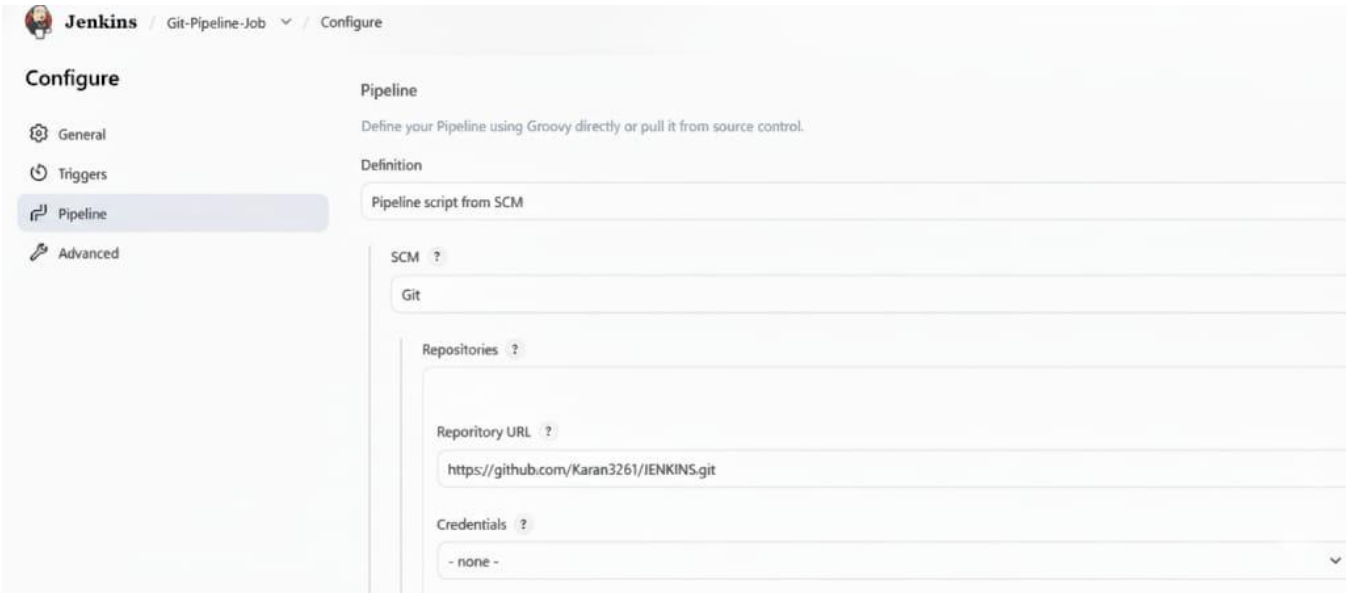


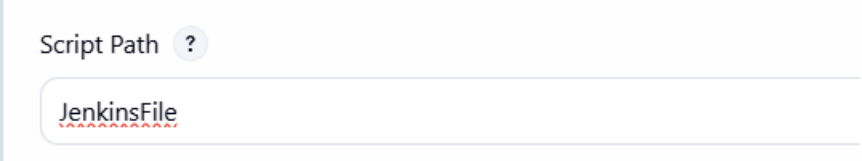
2.Commit Jenkinsfile to GitHub



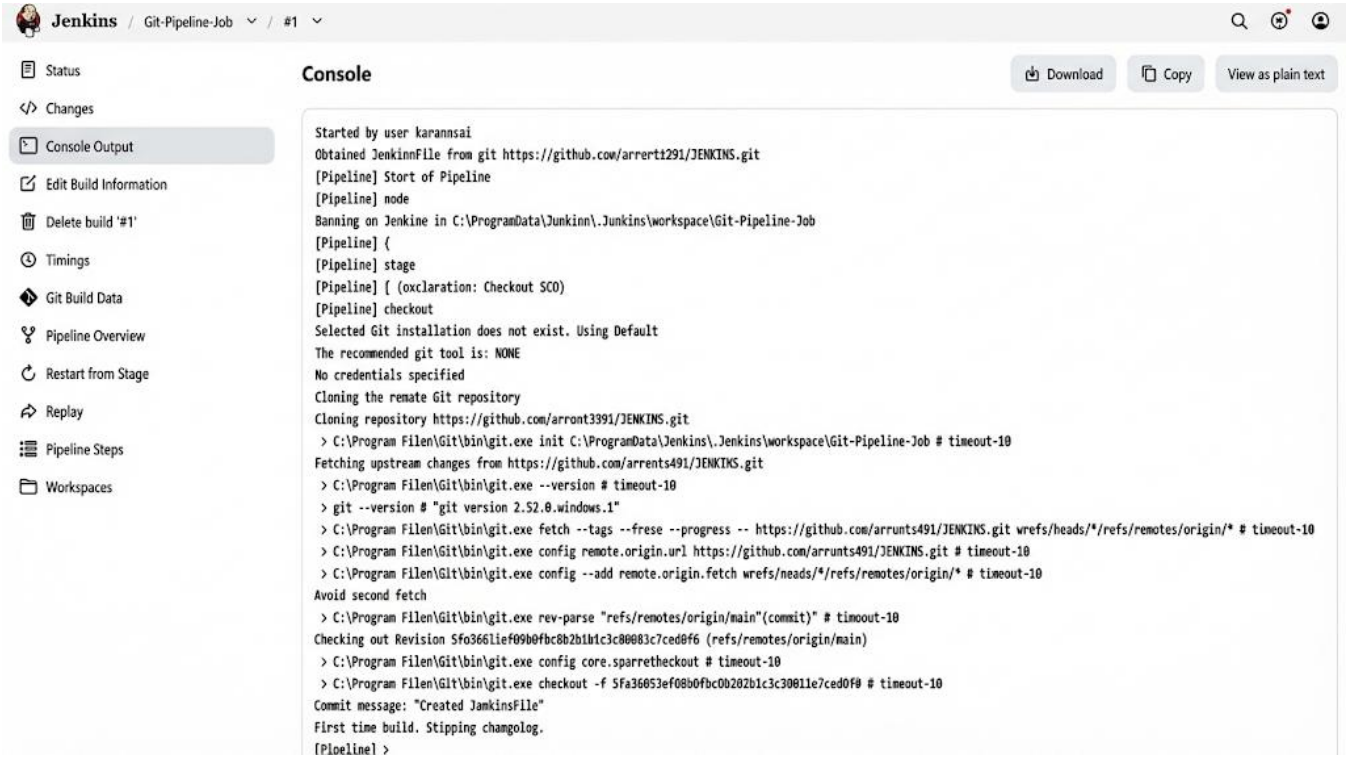
3.Configure Pipeline job:

SCM → Git





4.Trigger build automatically



```
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Build from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Test from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploy from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins / Git-Pipeline-Job

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

Builds >

Filter /

Today

#1 9:10 PM

Git-Pipeline-Job

Stage View

	Declarative: Checkout SCM	Build	Test	Deploy
Average stage times: (full run time: ~7s)	3s	138ms	115ms	110ms
#1 Feb 04 21:10 No Changes	3s	138ms	115ms	110ms

Permalinks

- Last build (#1), 8 min 46 sec ago
- Last stable build (#1), 8 min 46 sec ago
- Last successful build (#1), 8 min 46 sec ago
- Last completed build (#1), 8 min 46 sec ago

5.Modify Jenkinsfile and rebuild

JENKINS / JenkinsFile in main

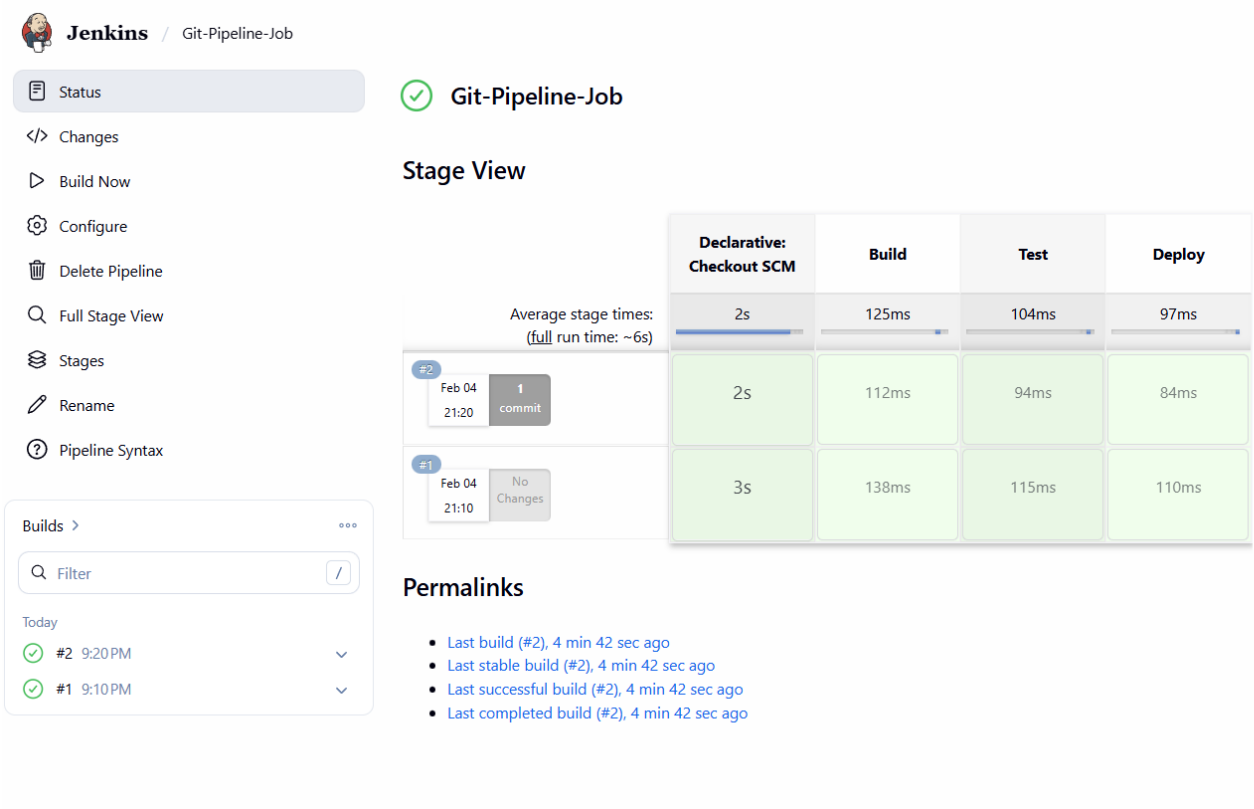
Edit Preview

```
pipeline {
  agent any
  stages {
    stage('Build') {
      steps {
        echo 'Build from Jenkinsfile,Build Stage Updated'
      }
    }
    stage('Test') {
      steps {
        echo 'Test from Jenkinsfile'
      }
    }
    stage('Deploy') {
      steps {
        echo 'Deploy from Jenkinsfile'
      }
    }
  }
}
```

karan326 Updated JenkinsFile ddabd7c · now 2 Commits

JenkinsFile Updated JenkinsFile now

UPDATED CONSOLE OUTPUT:

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Build from Jenkinsfile,Build Stage Updated
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Test from Jenkinsfile
```

6.View Stage View